



Completing the Square



A). Solve the equations.

E.g. $(f - 3)^2 = 9$, square root both sides,
 $f - 3 = \pm 3$,
 so $f = 3 + 3 = 6$ or $-3 + 3 = 0$.



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|-----------------------|-----------------------|-----------------------|-----------------------|
| 1). $x^2 = 64$ | 2). $d^2 = 100$ | 3). $x^2 = 49$ | 4). $h^2 = 10$ |
| 5). $(c - 2)^2 = 9$ | 6). $(h - 7)^2 = 4$ | 7). $(x + 3)^2 = 4$ | 8). $(p + 2)^2 = 25$ |
| 9). $(x - 1)^2 = 2$ | 10). $(q + 4)^2 = 3$ | 11). $(w - 3)^2 = 5$ | 12). $(a + 2)^2 = 2$ |
| 13). $(h - 6)^2 = 36$ | 14). $(c - 4)^2 = 10$ | 15). $(x + 3)^2 = 49$ | 16). $(m - 1)^2 = 7$ |
| 17). $(t - 8)^2 = 3$ | 18). $(y + 7)^2 = 6$ | 19). $(h + 9)^2 = 3$ | 20). $(c + 10)^2 = 8$ |

B). Expand the brackets. The expressions are all perfect squares.

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|------------------|-------------------|-----------------|-----------------|
| 1). $(x + 4)^2$ | 2). $(a - 6)^2$ | 3). $(r + 7)^2$ | 4). $(e - 5)^2$ |
| 5). $(p - 3)^2$ | 6). $(t + 2)^2$ | 7). $(b - 8)^2$ | 8). $(c - 9)^2$ |
| 9). $(k - 12)^2$ | 10). $(u + 11)^2$ | | |

C). Add the term that will make each expression a **perfect square**, then factorise it.

E.g. To find the number we halve the coefficient of x and then square it.

$$x^2 + 6x$$

Halve 6 and then square it, i.e. 9.

Therefore, we have $x^2 + 6x + 9 = (x + 3)^2$

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|------------------|---------------------------|---------------------------|------------------|
| 1). $a^2 + 8a$ | 2). $b^2 + 10b$ | 3). $c^2 - 4c$ | 4). $d^2 - 6d$ |
| 5). $x^2 + 5x$ | 6). $y^2 - 3y$ | 7). $z^2 - 7z$ | 8). $m^2 + 2m$ |
| 9). $n^2 - n$ | 10). $n^2 - 12n$ | 11). $x^2 + 9x$ | 12). $k^2 + 11k$ |
| 13). $b^2 - 13b$ | 14). $v^2 - \frac{1}{2}v$ | 15). $j^2 + \frac{1}{4}j$ | |

D). Solve these equations by completing the square. Some will factorise, use this as a check.

E.g. $x^2 - 8x + 3 = 0$

$$x^2 - 8x = -3$$

$$x^2 - 8x + 16 = 13$$

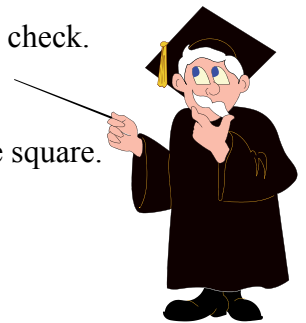
$$(x - 4)^2 = 13$$

Therefore, $x - 4 = \pm \sqrt{13}$

$$x = 4 \pm \sqrt{13}$$

Add 16 to both sides to complete the square.

$x = 0.39$ or $x = 7.61$ (2 d.p.)



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|---------------------------|--------------------------|--------------------------|
| 1). $a^2 + 4a - 21 = 0$ | 2). $b^2 - b - 12 = 0$ | 3). $n^2 + 4n + 4 = 0$ |
| 4). $d^2 - 5d + 6 = 0$ | 5). $g^2 + 5g + 4 = 0$ | 6). $x^2 - 10x + 25 = 0$ |
| 7). $q^2 + 10q + 22 = 0$ | 8). $t^2 - 6t + 9 = 0$ | 9). $m^2 + 6m + 7 = 0$ |
| 10). $y^2 - 3y + 1 = 0$ | 11). $d^2 + 2d - 2 = 0$ | 12). $x^2 - 4x - 2 = 0$ |
| 13). $k^2 - 5k + 2 = 0$ | 14). $f^2 - f - 1 = 0$ | 15). $x^2 + 3x - 2 = 0$ |
| 16). $p^2 - 10p + 15 = 0$ | 17). $v^2 + 9v + 19 = 0$ | 18). $u^2 - 14u - 3 = 0$ |

E). Solve these equations by completing the square. Some will factorise, use this as a check.

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|----------------------|-------------------------|----------------------|-----------------------|
| 1). $c^2 + 2c = 3$ | 2). $d^2 - 4d = 5$ | 3). $x^2 - 3x = -2$ | 4). $x^2 - 2x = 4$ |
| 5). $y^2 + 2y = 1$ | 6). $h^2 + 6h = 5$ | 7). $n^2 - 2n = 2$ | 8). $f^2 - 4f = -1$ |
| 9). $w^2 - 8w = -13$ | 10). $d^2 + 6d = -7$ | 11). $e^2 = 6e - 4$ | 12). $f^2 = 11 - 4f$ |
| 13). $j^2 = -4j - 2$ | 14). $n^2 = 2n + 1$ | 15). $h^2 = h + 5$ | 16). $d^2 = 12d - 35$ |
| 17). $h^2 + h = 8$ | 18). $e^2 - 6e - 3 = 0$ | 19). $x^2 + 5x = 15$ | 20). $e^2 = 3e + 11$ |